



REPORT ES Pilot



GENUKA GIYON

CircuitSeeker

ELECTRONIC FUNDAMENTALS

RESISTORS | CAPACITORS

Knowledge of concepts, working principles & equations that govern these vital components

BASIC ANALOG

TRANSISTORS | OP-AMPS

Knowledge of concepts, working principles & equations that govern these vital components

ICs & APPLICATIONS

LOGIC ICs | OP-AMP | 555 TIMER

Making useful circuit systems referring to datasheets without a microcontroller.

MCU

INTERRUPTS | EEPROM | MILLIS | PARALLEL PROCESSING

Practical know-how of how & when to use these more advanced features in microcontrollers.

CIRCUITRY

COMPONENT INTEGRATION | CIRCUIT CONNECTIONS | I/O

IDENTIFICATION

Understanding relevant sensor & actuator functions & physics. Using them effectively for their product through a microcontroller.

CODING

ARDUINO C PROGRAMMING | FUNCTIONS | LOOPS | LIBRARIES

Programming the microcontroller to give necessary outputs to the actuators according to the inputs from the sensors & realise the functions of the product.

PCB

CIRCUIT DIAGRAMS | SYMBOLS | PCB DESIGNING | PCB ETCHING | DRILLING | SOLDERING

Designing their own Printed Circuit Boards & printing through industry-standard suppliers.

DESIGN THINKING

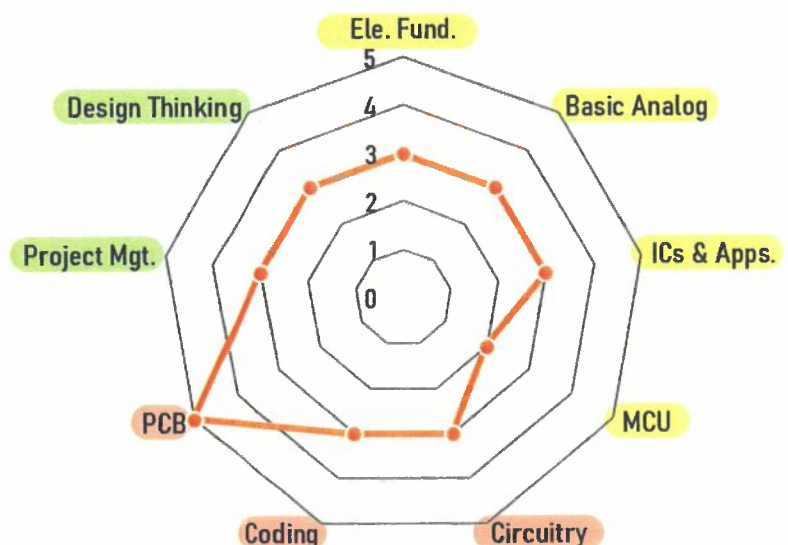
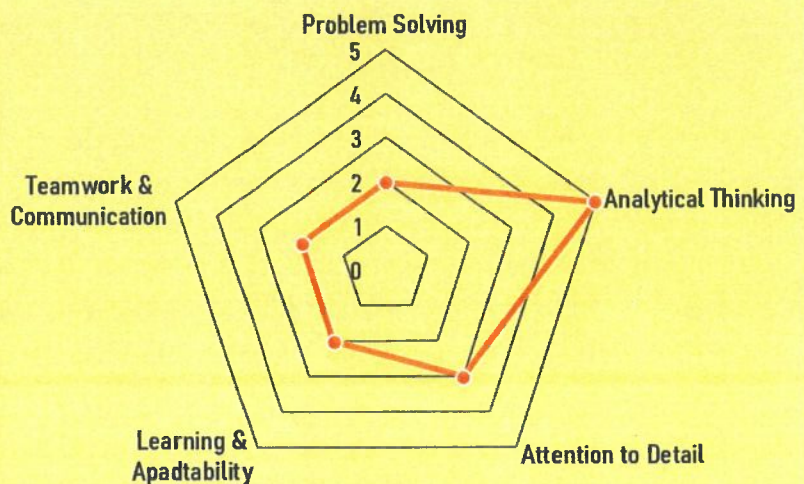
PRACTICAL MECHANICS | CHOICE OF MATERIALS | CAD MODEL

Come up with a practical design, complete CAD model of the final product for manufacturing.

PROJECT MANAGEMENT

PLANNING | WORK DISTRIBUTION | TIMELINES | EVALUATION

Making a work plan & sticking to it while making necessary changes after evaluation.





TEAM PROJECT: **OBJECT PICKING LINE FOLLOWER**

SKILLS LEARNED

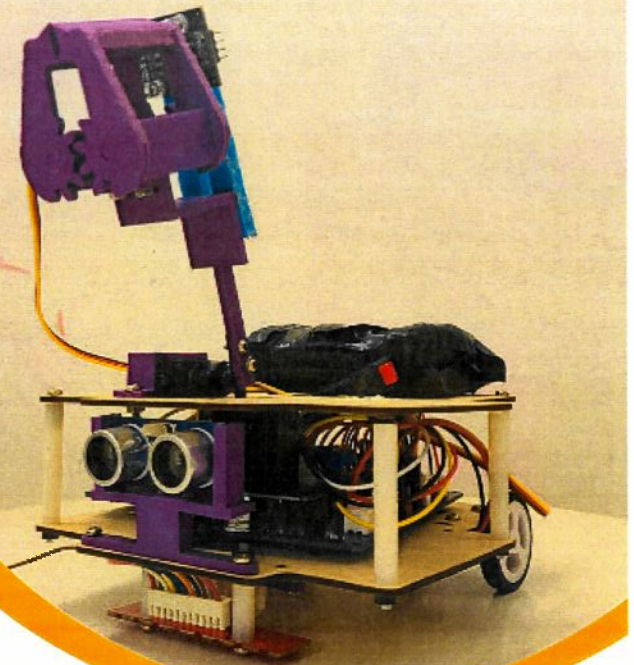
- ✓ Working with IR array using Arduino Nano.
- ✓ Working with Colour Sensor using Arduino Nano.
- ✓ Working with L293D Motor Control IC.
- ✓ PID Control Loops.
- ✓ MCU Pin Management.
- ✓ Getting Professionally manufactured PCBs.

TO BE IMPROVED

- ✓ Programming by referring to data sheets of components
- ✓ Independent error fixing
- ✓ Project Management
- ✓ Integrating & smooth operation of multiple modules.
- ✓ Modelling & Simulation at Ideation



FULL REPORT



PERSONAL PROFILE

Genuka, your ability to recall and apply prior knowledge is truly commendable and reflects your dedication to practicing theories and exploring additional activities. Your strong analytical thinking shines through, and it's impressive how you've managed to handle most of the project work independently.

While there are occasional challenges in practical tasks and a bit of hesitation with complex topics, your significant improvement is a testament to your resilience and determination. Keep pushing your boundaries and embracing new challenges—you have the potential to achieve even greater heights. Best of luck, and may your journey be as legendary as CR7's! Keep up the great work, Genuka!

ELIGIBLE PROGRAMMES:

EMBEDDED SYSTEMS PROGRAM



MEU LABS
LEARN IT DIFFERENT